

NSF 22-600: Joint DMS/NIGMS Initiative to Support Research at the Interface of the Biological and Mathematical Sciences (DMS/NIGMS)

Program Solicitation

Document Information

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National Science Foundation

Directorate for Mathematical and Physical Sciences
Division of Mathematical Sciences



National Institutes of Health

National Institute of General Medical Sciences

Submission Window Date(s) (due by 5 p.m. submitter's local time):

September 01, 2022 - September 19, 2022

September 1 - September 18, Annually Thereafter



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Important Information And Revision Notes

This solicitation was revised to include compliance with the NIGMS mission.

Innovating and migrating proposal preparation and submission capabilities from FastLane to Research.gov is part of the ongoing NSF information technology modernization efforts, as described in [Important Notice No. 147](#). In support of these efforts, proposals submitted in response to this program solicitation must be prepared and submitted via Research.gov or via Grants.gov and may not be prepared or submitted via FastLane.

Any proposal submitted in response to this solicitation should be submitted in accordance with the revised *NSF Proposal & Award Policies & Procedures Guide* (PAPPG) ([NSF 22-1](#)), which is effective for proposals submitted, or due, on or after October 4, 2021.

Summary Of Program Requirements

General Information

Program Title:

Joint DMS/NIGMS Initiative to Support Research at the Interface of the Biological and Mathematical Sciences (DMS/NIGMS)

Synopsis of Program:

The Division of Mathematical Sciences (DMS) in the Directorate for Mathematical and Physical Sciences (MPS) at the National Science Foundation (NSF) and the National Institute of General Medical Sciences (NIGMS) at the National Institutes of Health (NIH) plan to support fundamental research in mathematics and statistics necessary to answer questions in the biological and biomedical sciences. Both agencies recognize the need to promote research at the interface between mathematical and life sciences. This program is designed to encourage new collaborations, as well as to support innovative activities by existing teams.

The joint DMS/NIGMS initiative offers two submission tracks: Track 1 - for projects with a total budget of up to \$600,000 for an award duration of 3 years, and Track 2 - for projects with a total budget of up to \$1,200,000 for an award duration of 3-4 years.

Cognizant Program Officer(s):

Please note that the following information is current at the time of publishing. See program website for any updates to the points of contact.

- Zhilan J. Feng, Program Director, NSF/DMS, telephone: (703) 292-7523, email: dms-nigms@nsf.gov
- Vu Dinh, Program Director, NSF/DMS, telephone: (703) 292-4306, email: vdinh@nsf.gov
- John E. Kolassa, Program Director, telephone: (703) 292-4733, email: jkolassa@nsf.gov
- Tapabrata Maiti, Program Director, telephone: (703) 292-5307, email: tmaiti@nsf.gov
- Peggy Wang, Program Director, telephone: 301-594-0800, email: peggy.wang@nih.gov
- Han Nguyen, Program Director, NIH/NIGMS, telephone: (301) 496-7301, email: han.nguyen@nih.gov

Applicable Catalog of Federal Domestic Assistance (CFDA) Number(s):

- 47.049 --- Mathematical and Physical Sciences
- 93.859 --- National Institute of General Medical Sciences

Award Information

Anticipated Type of Award:

Standard Grant or Continuing Grant or R01 project (if the proposal is selected to be funded by NIGMS)

Estimated Number of Awards: 15 to 25

Approximately 15 to 25 awards from this competition may be made by either NSF or NIH at the option of the agencies, not the grantee. The number of awards will depend on the quality of received proposals and budget availability.

Anticipated Funding Amount: \$5,000,000

Up to \$5,000,000 per year for new proposals (up to \$2,000,000 from NSF and up to \$3,000,000 from NIGMS), subject to availability of funds and receipt of meritorious proposals. Track 1 – for exploratory projects of high-risk, high-reward, or those from new collaborative teams, with total budgets of up to \$600,000 for an award duration of 3 years. Track 2 - for projects of large scope from well-established teams, with total budgets of up to \$1,200,000 for an award duration of 3-4 years.

Eligibility Information

Who May Submit Proposals:

The categories of proposers eligible to submit proposals to the National Science Foundation are identified in the NSF Proposal & Award Policies & Procedures Guide (PAPPG), Chapter I.E. Unaffiliated individuals are not eligible to submit proposals in response to this solicitation.

Who May Serve as PI:

There are no restrictions or limits.

Limit on Number of Proposals per Organization:

There are no restrictions or limits.

Limit on Number of Proposals per PI or co-PI:

There are no restrictions or limits.

Proposal Preparation and Submission Instructions**A. Proposal Preparation Instructions**

- **Letters of Intent:** Not required
- **Preliminary Proposal Submission:** Not required
- **Full Proposals:**
 - Full Proposals submitted via Research.gov: *NSF Proposal and Award Policies and Procedures Guide* (PAPPG) guidelines apply. The complete text of the PAPPG is available electronically on the NSF website at: https://www.nsf.gov/publications/pub_summ.jsp?ods_key=pappg.
 - Full Proposals submitted via Grants.gov: *NSF Grants.gov Application Guide: A Guide for the Preparation and Submission of NSF Applications via Grants.gov* guidelines apply (Note: The *NSF Grants.gov Application Guide* is available on the Grants.gov website and on the NSF website at: https://www.nsf.gov/publications/pub_summ.jsp?ods_key=grantsgovguide).

B. Budgetary Information

- **Cost Sharing Requirements:**

Inclusion of voluntary committed cost sharing is prohibited.
- **Indirect Cost (F&A) Limitations:**

Not Applicable
- **Other Budgetary Limitations:**

Not Applicable

C. Due Dates

- **Submission Window Date(s)** (due by 5 p.m. submitter's local time):

September 01, 2022 - September 19, 2022

September 1 - September 18, Annually Thereafter

Proposal Review Information Criteria**Merit Review Criteria:**

National Science Board approved criteria. Additional merit review criteria apply. Please see the full text of this solicitation for further information.

Award Administration Information**Award Conditions:**

Additional award conditions apply. Please see the full text of this solicitation for further information.

Reporting Requirements:

Additional reporting requirements apply. Please see the full text of this solicitation for further information.

I. Introduction

The extraordinary growth of available data has created tremendous opportunities for revolutionary mathematics-driven advances in biological and biomedical research. In this initiative, the National Institute of General Medical Sciences (NIGMS) and the National Science Foundation's Division of Mathematical Sciences (NSF/DMS) partner to promote research at the interface of the biological, biomedical, and mathematical sciences. The expertise of DMS in mathematical, computational, and statistical sciences and the complementary expertise of NIGMS in biological and biomedical research are expected to provide support for novel and transformative quantitative life sciences research.

A successful proposal is expected to address clearly stated biological/biomedical questions, make a compelling case for and develop innovative mathematical/computational/statistical methods or integrate disparate mathematical/computational/statistical fields, and articulate a well-defined plan for the mathematics or statistics to drive biological/biomedical discovery within the funded period. A clear description on the contribution of the mathematical/computational/statistical work to improved understanding of biological/biomedical problems is required. Research collaborations that include scientists from both the life and mathematical science communities are expected. Proposals from single investigators must provide convincing evidence that the principal investigator has the necessary expertise in both mathematics/statistics and life sciences. While projects from existing collaborations are allowed, we strongly encourage exploratory, high-risk and high-reward Track 1 proposals from new teams of researchers.

All proposals in response to this solicitation must address important questions in the biological and biomedical sciences that are within the NIGMS mission (<https://www.nigms.nih.gov/research-areas>), which is to support fundamental research that increases our understanding of basic biological processes and lays the foundation for future advances in disease diagnosis, treatment, and prevention. NIGMS supports research and technology development that is aimed at understanding general principles, mechanisms, and processes through Scientific Divisions with diverse interests. NIGMS does not focus on single classes of cells, tissues, organs, or diseases unless they are used as models for elucidating basic principles. (<http://www.nigms.nih.gov/about/Pages/contactbyarea.aspx>). Proposals that do not conform to the NIGMS mission will be returned without review.

Note: Proposals submitted to this solicitation for work based on topics or approaches for which the PI, co-PI or Senior Personnel previously received support from the joint DMS/NIGMS program will be returned without review.

II. Program Description

The Division of Mathematical Sciences (DMS) within the Directorate for Mathematical and Physical Sciences (MPS) and the National Institute of General Medical Sciences (NIGMS) anticipate supporting innovative research in the mathematical and statistical sciences motivated by biological and biomedical applications. Appropriate application areas of biological and biomedical sciences are those currently supported by the National Institute of General Medical Sciences (cf., <https://www.nigms.nih.gov/research-areas>; <https://www.nigms.nih.gov/Research/specificareas/mathbio>).

Investigators are strongly encouraged to contact the Management Team via email (DMS-NIGMS@nsf.gov) before submitting a proposal to discuss suitability. Email inquiries should (i) identify the key personnel (PI, co-PI, and Senior Personnel) and their research expertise, (ii) provide a brief description of the biological/biomedical application and the novelty of the proposed quantitative approach.

Mathematical scientists (pure, applied, computational, and/or statistical) and others capable of developing innovative methods and tools in mathematical sciences, are encouraged to apply. Successful projects to the joint DMS/NIGMS initiative are anticipated to impact and advance biological/biomedical research and lead to advances in the mathematical sciences. Interdisciplinary collaborations between mathematical/statistical and biological/biomedical scientists are expected and highly preferred.

Examples of relevant research areas/topics of inquiry to the joint DMS/NIGMS initiative include, but are not limited to:

- New mathematical, computational, and statistical methods that address the efficiency, robustness, quality control, uncertainty qualification, and reproducibility in biological and biomedical research

- Mechanistic and/or first principle-based modeling of biological/biomedical systems with novel mathematical or computational approaches at all levels of biological systems (e.g., molecular, cellular, organismal, and population levels)
- Novel and foundational mathematical/computational/statistical approaches for data science research of complex biological/biomedical systems (e.g., unique data analysis approaches, new approaches for integrating multiple data types, data imputation, interpretable AI/ML/deep learning models, model validation, methods for model evaluation, transferable models, novel approaches for data cataloging and structuring for use in analysis)
- Parameter estimation, inference and uncertainty quantification for multi-scale deterministic and/or stochastic models for complex biological processes, model validation and robustness evaluation
- Predictive models from high-dimensional or sparse biological/biomedical data with uncertainty
- Integrated multiscale approaches for connecting multiple spatiotemporal scales of biological processes

Special formatting and submission requirements for proposals to the joint DMS/NIGMS initiative are described under Proposal Preparation and Submission Instructions. Pls are strongly encouraged to follow these guidelines carefully; proposals that do not strictly adhere to requirements will be returned without review.

This program provides two tracks that support projects of different scale and scope:

- DMS/NIGMS Track 1: This track provides support for projects with a total budget (including indirect costs) of up to \$600,000 for an award duration of up to 3 years. Successful Track 1 projects are anticipated to be high-risk, high-reward exploratory projects, or those from new teams of collaborators with complementary research expertise.
- DMS/NIGMS Track 2: This track provides support for projects with a total budget (including indirect costs) of up to \$1,200,000 for an award duration of 3-4 years.

This joint program will not fund foreign institutions. International collaborators are welcome to participate in the projects, but they must obtain support from their respective international funding organizations. Certain expenses, such as expenses which occur during short-term visits may be covered by the US host institution. However, these expenses must be explained in the budget discussion and be justified. Evidence of support for the foreign component of the proposed research must be provided in the proposal.

III. Award Information

Anticipated Type of Award: Standard Grant or Continuing Grant or R01 project (if the proposal is selected to be funded by NIGMS)

Estimated Number of Awards: 15 to 25

Approximately 15 to 25 awards from this competition may be made by either NSF or NIH at the option of the agencies, not the grantee. The number of awards will depend on the quality of received proposals and budget availability.

Anticipated Funding Amount: \$5,000,000

Up to \$5,000,000 per year for new proposals (up to \$2,000,000 from NSF and up to \$3,000,000 from NIGMS), subject to availability of funds and receipt of meritorious proposals. Track 1 – for exploratory projects of high-risk, high-reward, or those from new collaborative teams, with total budgets of up to \$600,000 for an award duration of 3 years. Track 2 - for projects of large scope from well-established teams, with total budgets of up to \$1,200,000 for an award duration of 3-4 years.

Upon conclusion of the review process, meritorious proposals may be recommended for funding by either NIGMS or NSF, at the option of the agencies, not the grantee. Subsequent grant administration procedures will be in accordance with the individual policies of the awarding agency.

IV. Eligibility Information

Who May Submit Proposals:

The categories of proposers eligible to submit proposals to the National Science Foundation are identified in the NSF Proposal & Award Policies & Procedures Guide (PAPPG), Chapter I.E. Unaffiliated individuals are not eligible to submit proposals in response to this solicitation.

Who May Serve as PI:

There are no restrictions or limits.

Limit on Number of Proposals per Organization:

There are no restrictions or limits.

Limit on Number of Proposals per PI or co-PI:

There are no restrictions or limits.

V. Proposal Preparation And Submission Instructions

A. Proposal Preparation Instructions

Full Proposal Preparation Instructions: Proposers may opt to submit proposals in response to this Program Solicitation via Research.gov or Grants.gov.

- Full Proposals submitted via Research.gov: Proposals submitted in response to this program solicitation should be prepared and submitted in accordance with the general guidelines contained in the *NSF Proposal and Award Policies and Procedures Guide* (PAPPG). The complete text of the PAPPG is available electronically on the NSF website at: https://www.nsf.gov/publications/pub_summ.jsp?ods_key=pappg. Paper copies of the PAPPG may be obtained from the NSF Publications Clearinghouse, telephone (703) 292-8134 or by e-mail from nsfpubs@nsf.gov. The Prepare New Proposal setup will prompt you for the program solicitation number.
- Full proposals submitted via Grants.gov: Proposals submitted in response to this program solicitation via Grants.gov should be prepared and submitted in accordance with the *NSF Grants.gov Application Guide: A Guide for the Preparation and Submission of NSF Applications via Grants.gov*. The complete text of the *NSF Grants.gov Application Guide* is available on the Grants.gov website and on the NSF website at: https://www.nsf.gov/publications/pub_summ.jsp?ods_key=grantsgovguide). To obtain copies of the Application Guide and Application Forms Package, click on the Apply tab on the Grants.gov site, then click on the Apply Step 1: Download a Grant Application Package and Application Instructions link and enter the funding opportunity number, (the program solicitation number without the NSF prefix) and press the Download Package button. Paper copies of the Grants.gov Application Guide also may be obtained from the NSF Publications Clearinghouse, telephone (703) 292-8134 or by e-mail from nsfpubs@nsf.gov.

In determining which method to utilize in the electronic preparation and submission of the proposal, please note the following:

Collaborative Proposals. All collaborative proposals submitted as separate submissions from multiple organizations must be submitted via Research.gov. PAPPG Chapter II.D.3 provides additional information on collaborative proposals.

See PAPPG Chapter II.C.2 for guidance on the required sections of a full research proposal submitted to NSF. Please note that the proposal preparation instructions provided in this program solicitation may deviate from the PAPPG instructions.

The following instructions deviate from guidelines in the PAPPG and NSF Grants.gov Application Guide or emphasize changes from the last solicitation.

Prospective investigators are strongly encouraged to discuss their proposals with the program directors via email (DMS-NIGMS@nsf.gov) before submission to ask if the proposed project is relevant to both DMS and NIGMS. Biological

application areas are restricted to those currently supported by the NIGMS (see <http://www.nigms.nih.gov/Research/>); projects that do not conform to the NIGMS mission will be returned without review.

Proposal Title: Titles of Track 1 proposals submitted to the joint DMS/NIGMS initiative should begin with "DMS/NIGMS 1:" followed by a substantive title. Titles of Track 2 proposals should begin with "DMS/NIGMS 2:" followed by a substantive title.

Project Description: The 15-page project description **MUST** consist of two distinct parts: (1) A **maximum of 12 pages** at the beginning of the project description addressing the NSF criterion of *Intellectual Merit*. (Note that this NSF criterion corresponds with the NIH criteria of Significance, Investigators, Innovation, Approach, and Environment; see "Additional Solicitation Specific Review Criteria" below); and (2) a separate section at the end of the project description, **not more than 3 pages long**, clearly titled "**Broader Impacts**," which must address the NSF criterion of *Broader Impacts* (see "Merit Review Criteria" below for a full definition and discussion of what constitutes Broader Impacts).

Results from Prior NSF and/or NIH Support: If any Principal Investigator (PI) or co-PI identified on the project has received NSF or NIH funding with (i) an award with an end date in the past five years, or (ii) any current funding, including any no cost extensions, information on the award(s) **is required**. Each PI and co-PI who has received more than one award must report on the award most closely related to the proposal. No **more than five total** pages may be used to describe the results, which must be summarized in a **single, separate section, clearly titled "Results from Prior NSF/NIH Support."** This section may be included in either the 12-pages covering Intellectual Merit or the 3-pages on Broader Impacts, at the PI's discretion.

The following information must be provided:

- a. *the NSF or NIH award number, amount and period of support;*
- b. *the title of the project;*
- c. *a summary of the results of the completed work, including accomplishments related to the Broader Impact activities supported by the award and, for a research project, any contribution to the development of human resources in science and engineering;*
- d. *publications resulting from the NSF or NIH award; and*
- e. *evidence of research products and their availability, including, but not limited to: data, publications, samples, physical collections, software, and models, as described in any Data Management Plan.*

Reviewers will be asked to comment on the quality of the prior work described in this section of the proposal.

Project Management Plan: All proposals **MUST** include a Project Management Plan, **no more than 3 pages long**, submitted as a **separate Supplementary Document**.

1. For projects involving more than one investigator, the plan should discuss the role of each investigator, as well as the governance and organizational structure of the leadership team and the research project, including communication plans, process for making decisions on scientific direction, and procedures for resolving conflicts.
2. For projects with only one investigator, the plan should document the PI's necessary qualifications in both biological/biomedical and mathematical sciences.

The roles and administrative, technical, and scientific responsibilities for the project should be delineated for the investigator(s) and other collaborators. The distribution of resources to specific components of the project must be delineated in the Project Management Plan. In the event of an award, the requested allocation may be reflected in a footnote on the NIH Notice of Grant Award (NOGA). This information should be submitted separately as a **Supplementary Document**.

Protection of Human Subjects/Use and Care of Vertebrate Animals: Both NSF and NIH have rules regarding the use of human subjects and/or vertebrate animals in research. Proposals that involve human subjects or use vertebrate animals **MUST INCLUDE** the information required by both agencies. See the NSF PAPPG AND the NIH SF 424 application guide for additional information (<https://grants.nih.gov/grants/how-to-apply-application-guide/forms-g/general/g.500-phs->

[human-subjects-and-clinical-trials-information.htm](#)). Information on the use of human subjects and/or vertebrate animals is considered in the review of the proposals and should be submitted separately as a **Supplementary Document**.

Letters of Collaboration: Letters of collaboration document significant collaborative arrangements and must be limited to stating the intent to collaborate and should not contain endorsements or evaluation of the proposed project or investigators. The recommended format for letters of collaboration is as follows:

"If the proposal submitted by Dr. [insert the full name of the Principal Investigator] entitled [insert the proposal title] is selected for funding by NSF or NIH, it is my intent to collaborate and/or commit resources as detailed in the Project Description or the Facilities, Equipment or Other Resources section of the proposal."

The Project Description should document the need for and nature of collaborations, such as intellectual contributions to the project, permission to access a site, an instrument, or a facility, offer of data, samples and materials for research, logistical support to the research and education program, or mentoring of U.S. students at a foreign site. All letters of collaboration must be included at the time of submission as separate **Supplementary Documents**.

Departure from this format may result in the proposal being returned without review.

B. Budgetary Information

Cost Sharing:

Inclusion of voluntary committed cost sharing is prohibited.

C. Due Dates

- **Submission Window Date(s)** (due by 5 p.m. submitter's local time):

September 01, 2022 - September 19, 2022

September 1 - September 18, Annually Thereafter

D. Research.gov/Grants.gov Requirements

For Proposals Submitted Via Research.gov:

To prepare and submit a proposal via Research.gov, see detailed technical instructions available at:

https://www.research.gov/research-portal/appmanager/base/desktop?_nfpb=true&_pageLabel=research_node_display&_nodePath=/researchGov/Service/Desktop/ProposalPreparationandSubmission.html. For Research.gov user support, call the Research.gov Help Desk at 1-800-673-6188 or e-mail rgov@nsf.gov. The Research.gov Help Desk answers general technical questions related to the use of the Research.gov system. Specific questions related to this program solicitation should be referred to the NSF program staff contact(s) listed in Section VIII of this funding opportunity.

For Proposals Submitted Via Grants.gov:

Before using Grants.gov for the first time, each organization must register to create an institutional profile. Once registered, the applicant's organization can then apply for any federal grant on the Grants.gov website. Comprehensive information about using Grants.gov is available on the Grants.gov Applicant Resources webpage: <https://www.grants.gov/web/grants/applicants.html>. In addition, the NSF Grants.gov Application Guide (see link in Section V.A) provides instructions regarding the technical preparation of proposals via Grants.gov. For Grants.gov user support, contact the Grants.gov Contact Center at 1-800-518-4726 or by email: support@grants.gov. The Grants.gov Contact Center answers general technical questions related to the use of Grants.gov. Specific questions related to this program solicitation should be referred to the NSF program staff contact(s) listed in Section VIII of this solicitation.

Submitting the Proposal: Once all documents have been completed, the Authorized Organizational Representative (AOR) must submit the application to Grants.gov and verify the desired funding opportunity and agency to which

the application is submitted. The AOR must then sign and submit the application to Grants.gov. The completed application will be transferred to the NSF FastLane system for further processing.

Proposers that submitted via Research.gov may use Research.gov to verify the status of their submission to NSF. For proposers that submitted via Grants.gov, until an application has been received and validated by NSF, the Authorized Organizational Representative may check the status of an application on Grants.gov. After proposers have received an e-mail notification from NSF, Research.gov should be used to check the status of an application.

VI. NSF Proposal Processing And Review Procedures

Proposals received by NSF are assigned to the appropriate NSF program for acknowledgement and, if they meet NSF requirements, for review. All proposals are carefully reviewed by a scientist, engineer, or educator serving as an NSF Program Officer, and usually by three to ten other persons outside NSF either as *ad hoc* reviewers, panelists, or both, who are experts in the particular fields represented by the proposal. These reviewers are selected by Program Officers charged with oversight of the review process. Proposers are invited to suggest names of persons they believe are especially well qualified to review the proposal and/or persons they would prefer not review the proposal. These suggestions may serve as one source in the reviewer selection process at the Program Officer's discretion. Submission of such names, however, is optional. Care is taken to ensure that reviewers have no conflicts of interest with the proposal. In addition, Program Officers may obtain comments from site visits before recommending final action on proposals. Senior NSF staff further review recommendations for awards. A flowchart that depicts the entire NSF proposal and award process (and associated timeline) is included in PAPPG Exhibit III-1.

A comprehensive description of the Foundation's merit review process is available on the NSF website at: https://www.nsf.gov/bfa/dias/policy/merit_review/.

Proposers should also be aware of core strategies that are essential to the fulfillment of NSF's mission, as articulated in [*Leading the World in Discovery and Innovation, STEM Talent Development and the Delivery of Benefits from Research - NSF Strategic Plan for Fiscal Years \(FY\) 2022 - 2026*](#). These strategies are integrated in the program planning and implementation process, of which proposal review is one part. NSF's mission is particularly well-implemented through the integration of research and education and broadening participation in NSF programs, projects, and activities.

One of the strategic objectives in support of NSF's mission is to foster integration of research and education through the programs, projects, and activities it supports at academic and research institutions. These institutions must recruit, train, and prepare a diverse STEM workforce to advance the frontiers of science and participate in the U.S. technology-based economy. NSF's contribution to the national innovation ecosystem is to provide cutting-edge research under the guidance of the Nation's most creative scientists and engineers. NSF also supports development of a strong science, technology, engineering, and mathematics (STEM) workforce by investing in building the knowledge that informs improvements in STEM teaching and learning.

NSF's mission calls for the broadening of opportunities and expanding participation of groups, institutions, and geographic regions that are underrepresented in STEM disciplines, which is essential to the health and vitality of science and engineering. NSF is committed to this principle of diversity and deems it central to the programs, projects, and activities it considers and supports.

A. Merit Review Principles and Criteria

The National Science Foundation strives to invest in a robust and diverse portfolio of projects that creates new knowledge and enables breakthroughs in understanding across all areas of science and engineering research and education. To identify which projects to support, NSF relies on a merit review process that incorporates consideration of both the technical aspects of a proposed project and its potential to contribute more broadly to advancing NSF's mission "to promote the progress of science; to advance the national health, prosperity, and welfare; to secure the national defense; and for other purposes." NSF makes every effort to conduct a fair, competitive, transparent merit review process for the selection of projects.

1. Merit Review Principles

These principles are to be given due diligence by PIs and organizations when preparing proposals and managing projects, by reviewers when reading and evaluating proposals, and by NSF program staff when determining whether or not to recommend proposals for funding and while overseeing awards. Given that NSF is the primary federal agency charged with nurturing and supporting excellence in basic research and education, the following three principles apply:

- All NSF projects should be of the highest quality and have the potential to advance, if not transform, the frontiers of knowledge.
- NSF projects, in the aggregate, should contribute more broadly to achieving societal goals. These "Broader Impacts" may be accomplished through the research itself, through activities that are directly related to specific research projects, or through activities that are supported by, but are complementary to, the project. The project activities may be based on previously established and/or innovative methods and approaches, but in either case must be well justified.
- Meaningful assessment and evaluation of NSF funded projects should be based on appropriate metrics, keeping in mind the likely correlation between the effect of broader impacts and the resources provided to implement projects. If the size of the activity is limited, evaluation of that activity in isolation is not likely to be meaningful. Thus, assessing the effectiveness of these activities may best be done at a higher, more aggregated, level than the individual project.

With respect to the third principle, even if assessment of Broader Impacts outcomes for particular projects is done at an aggregated level, PIs are expected to be accountable for carrying out the activities described in the funded project. Thus, individual projects should include clearly stated goals, specific descriptions of the activities that the PI intends to do, and a plan in place to document the outputs of those activities.

These three merit review principles provide the basis for the merit review criteria, as well as a context within which the users of the criteria can better understand their intent.

2. Merit Review Criteria

All NSF proposals are evaluated through use of the two National Science Board approved merit review criteria. In some instances, however, NSF will employ additional criteria as required to highlight the specific objectives of certain programs and activities.

The two merit review criteria are listed below. **Both** criteria are to be given **full consideration** during the review and decision-making processes; each criterion is necessary but neither, by itself, is sufficient. Therefore, proposers must fully address both criteria. (PAPPG Chapter II.C.2.d(i). contains additional information for use by proposers in development of the Project Description section of the proposal). Reviewers are strongly encouraged to review the criteria, including PAPPG Chapter II.C.2.d(i), prior to the review of a proposal.

When evaluating NSF proposals, reviewers will be asked to consider what the proposers want to do, why they want to do it, how they plan to do it, how they will know if they succeed, and what benefits could accrue if the project is successful. These issues apply both to the technical aspects of the proposal and the way in which the project may make broader contributions. To that end, reviewers will be asked to evaluate all proposals against two criteria:

- **Intellectual Merit:** The Intellectual Merit criterion encompasses the potential to advance knowledge; and
- **Broader Impacts:** The Broader Impacts criterion encompasses the potential to benefit society and contribute to the achievement of specific, desired societal outcomes.

The following elements should be considered in the review for both criteria:

1. What is the potential for the proposed activity to
 - a. Advance knowledge and understanding within its own field or across different fields (Intellectual Merit); and
 - b. Benefit society or advance desired societal outcomes (Broader Impacts)?

2. To what extent do the proposed activities suggest and explore creative, original, or potentially transformative concepts?
3. Is the plan for carrying out the proposed activities well-reasoned, well-organized, and based on a sound rationale? Does the plan incorporate a mechanism to assess success?
4. How well qualified is the individual, team, or organization to conduct the proposed activities?
5. Are there adequate resources available to the PI (either at the home organization or through collaborations) to carry out the proposed activities?

Broader impacts may be accomplished through the research itself, through the activities that are directly related to specific research projects, or through activities that are supported by, but are complementary to, the project. NSF values the advancement of scientific knowledge and activities that contribute to achievement of societally relevant outcomes. Such outcomes include, but are not limited to: full participation of women, persons with disabilities, and other underrepresented groups in science, technology, engineering, and mathematics (STEM); improved STEM education and educator development at any level; increased public scientific literacy and public engagement with science and technology; improved well-being of individuals in society; development of a diverse, globally competitive STEM workforce; increased partnerships between academia, industry, and others; improved national security; increased economic competitiveness of the United States; and enhanced infrastructure for research and education.

Proposers are reminded that reviewers will also be asked to review the Data Management Plan and the Postdoctoral Researcher Mentoring Plan, as appropriate.

Additional Solicitation Specific Review Criteria

Proposals submitted to this competition will be evaluated based on their value in advancing mathematical or statistical theory or methodology, as well as their impact on important biological/biomedical problems. Both NIH and NSF review criteria will be used. When responding to the NIH review criteria, reviewers will provide an overall NIH impact score to reflect their assessment of the likelihood for the project to exert a sustained, powerful influence on the research field(s) involved, in consideration of the following five scored review criteria, and additional review criteria. Reviewers will consider each of the review criteria below in the determination of scientific merit. An application does not need to be strong in all categories to be judged likely to have major scientific impact. The five core review criteria for NIH are:

- **Significance:** Does the project address an important problem or a critical barrier to progress in the field? Is the prior research that serves as the key support for the proposed project rigorous? If the aims of the project are achieved, how will scientific knowledge, technical capability, and/or clinical practice be improved? How will successful completion of the aims change the concepts, methods, technologies, treatments, services, or preventative interventions that drive this field?
- **Investigator(s):** Are the Principal Investigators, collaborators, and other researchers well suited to the project? If Early Stage Investigators or New Investigators, or in the early stages of independent careers, do they have appropriate experience and training? If established, have they demonstrated an ongoing record of accomplishments that have advanced their field(s)? If the project involves multiple investigators, do the investigators have complementary and integrated expertise; are their leadership approach, governance and organizational structure appropriate for the project?
- **Innovation:** Does the application challenge and seek to shift current research or clinical practice paradigms by utilizing novel theoretical concepts, approaches or methodologies, instrumentation, or interventions? Are the concepts, approaches or methodologies, instrumentation, or interventions novel to one field of research or novel in a broad sense? Is a refinement, improvement, or new application of theoretical concepts, approaches or methodologies, instrumentation, or interventions proposed?
- **Approach:** Are the overall strategy, methodology, and analyses well-reasoned and appropriate to accomplish the specific aims of the project? Have the investigators included plans to address weaknesses in the rigor of prior research that serves as the key support for the proposed project? Are potential problems, alternative strategies, and benchmarks for success presented? If the project is in the early stages of development, will the strategy establish feasibility and will particularly risky aspects be managed? Have the investigators presented adequate

plans to address relevant biological variables, such as sex, for studies in vertebrate animals or human subjects? If the project involves human subjects and/or NIH-defined clinical research, are the plans to address 1) the protection of human subjects from research risks, and 2) inclusion justified in terms of the scientific goals and research strategy proposed?

- Environment: Will the scientific environment in which the work will be done contribute to the probability of success? Are the institutional support, equipment, and other physical resources available to the investigators adequate for the project proposed? Will the project benefit from unique features of the scientific environment, subject populations, or collaborative arrangements?

In addition to the above review criteria, the following criteria will be addressed and considered in the determination of scientific merit and the rating if any are relevant to the proposed project:

- Protections for Human Subjects: For research that involves human subjects but does not involve one of the categories of research that are exempt under 45 CFR Part 46, the committee will evaluate the justification for involvement of human subjects and the proposed protections from research risk relating to their participation according to the following five review criteria: 1) risk to subjects, 2) adequacy of protection against risks, 3) potential benefits to the subjects and others, 4) importance of the knowledge to be gained, and 5) data and safety monitoring for clinical trials. For research that involves human subjects and meets the criteria for one or more of the categories of research that are exempt under 45 CFR Part 46, the committee will evaluate: 1) the justification for the exemption, 2) human subjects involvement and characteristics, and 3) sources of materials. For additional information on review of the Human Subjects section, please refer to the NIH [Guidelines for the Review of Human Subjects](#).
- Inclusion of Human Subjects Policies: When the proposed project involves human subjects and/or NIH-defined clinical research, the committee will evaluate the proposed plans for inclusion. For additional information on review of the Inclusion section, please refer to the NIH [Guidelines for the Review of Inclusion in Clinical Research](#).
- Vertebrate Animals: The committee will evaluate the involvement of live vertebrate animals as part of the scientific assessment according to the following criteria: (1) description of proposed procedures involving animals, including species, strains, ages, sex, and total number to be used; (2) justifications for the use of animals versus alternative models and for the appropriateness of the species proposed; (3) interventions to minimize discomfort, distress, pain and injury; and (4) justification for euthanasia method if NOT consistent with the AVMA Guidelines for the Euthanasia of Animals. Reviewers will assess the use of chimpanzees as they would any other application proposing the use of vertebrate animals. For additional information on review of the Vertebrate Animals section, please refer to the NIH [Worksheet for Review of the Vertebrate Animal Section](#).
- Biohazards: If materials or procedures are proposed that are potentially hazardous to research personnel and/or the environment, the adequacy of the proposed protection will be assessed.

As applicable for the project proposed, reviewers will address each of the following review considerations, but will not consider them in providing an overall NIH impact score.

- Budget and Period Support: Reviewers will consider whether the budget and the requested period of support are fully justified and reasonable in relation to the proposed research.
- Additional Comments to the Applicant: Reviewers may provide guidance to the applicant or recommend against resubmission without fundamental revision.

B. Review and Selection Process

Proposals submitted in response to this program solicitation will be reviewed by Ad hoc Review and/or Panel Review.

The review will be conducted jointly by NSF and NIH. Program staff from NIGMS are given access to every proposal submitted to the Joint DMS/NIGMS Initiative, as well as to all review information. Awards may be made by either NSF or NIH, at the option of the agencies, not the grantee. DMS and NIGMS will jointly select research projects for funding from among those designated as recommended for funding by the review process. Following this decision, the proposals will be divided between the two agencies for funding based on mutually agreed criteria and availability of funds. Proposals to

be considered for funding by NIGMS will require the submission of an NIH R01 research grant application by the applicant organization. Those grant applications to be recommended for award by NIGMS will be subject to a second level of review by the National Advisory General Medical Sciences Council.

Reviewers will be asked to evaluate proposals using two National Science Board approved merit review criteria and, if applicable, additional program specific criteria. A summary rating and accompanying narrative will generally be completed and submitted by each reviewer and/or panel. The Program Officer assigned to manage the proposal's review will consider the advice of reviewers and will formulate a recommendation.

After scientific, technical and programmatic review and consideration of appropriate factors, the NSF Program Officer recommends to the cognizant Division Director whether the proposal should be declined or recommended for award. NSF strives to be able to tell applicants whether their proposals have been declined or recommended for funding within six months. Large or particularly complex proposals or proposals from new awardees may require additional review and processing time. The time interval begins on the deadline or target date, or receipt date, whichever is later. The interval ends when the Division Director acts upon the Program Officer's recommendation.

After programmatic approval has been obtained, the proposals recommended for funding will be forwarded to the Division of Grants and Agreements or the Division of Acquisition and Cooperative Support for review of business, financial, and policy implications. After an administrative review has occurred, Grants and Agreements Officers perform the processing and issuance of a grant or other agreement. Proposers are cautioned that only a Grants and Agreements Officer may make commitments, obligations or awards on behalf of NSF or authorize the expenditure of funds. No commitment on the part of NSF should be inferred from technical or budgetary discussions with a NSF Program Officer. A Principal Investigator or organization that makes financial or personnel commitments in the absence of a grant or cooperative agreement signed by the NSF Grants and Agreements Officer does so at their own risk.

Once an award or declination decision has been made, Principal Investigators are provided feedback about their proposals. In all cases, reviews are treated as confidential documents. Verbatim copies of reviews, excluding the names of the reviewers or any reviewer-identifying information, are sent to the Principal Investigator/Project Director by the Program Officer. In addition, the proposer will receive an explanation of the decision to award or decline funding.

VII. Award Administration Information

A. Notification of the Award

Notification of the award is made to *the submitting organization* by an NSF Grants and Agreements Officer. Organizations whose proposals are declined will be advised as promptly as possible by the cognizant NSF Program administering the program. Verbatim copies of reviews, not including the identity of the reviewer, will be provided automatically to the Principal Investigator. (See Section VI.B. for additional information on the review process.)

B. Award Conditions

An NSF award consists of: (1) the award notice, which includes any special provisions applicable to the award and any numbered amendments thereto; (2) the budget, which indicates the amounts, by categories of expense, on which NSF has based its support (or otherwise communicates any specific approvals or disapprovals of proposed expenditures); (3) the proposal referenced in the award notice; (4) the applicable award conditions, such as Grant General Conditions (GC-1)*; or Research Terms and Conditions* and (5) any announcement or other NSF issuance that may be incorporated by reference in the award notice. Cooperative agreements also are administered in accordance with NSF Cooperative Agreement Financial and Administrative Terms and Conditions (CA-FATC) and the applicable Programmatic Terms and Conditions. NSF awards are electronically signed by an NSF Grants and Agreements Officer and transmitted electronically to the organization via e-mail.

*These documents may be accessed electronically on NSF's Website at https://www.nsf.gov/awards/managing/award_conditions.jsp?org=NSF. Paper copies may be obtained from the NSF Publications Clearinghouse, telephone (703) 292-8134 or by e-mail from nsfpubs@nsf.gov.

More comprehensive information on NSF Award Conditions and other important information on the administration of NSF awards is contained in the *NSF Proposal & Award Policies & Procedures Guide* (PAPPG) Chapter VII, available electronically on the NSF Website at https://www.nsf.gov/publications/pub_summ.jsp?ods_key=pappg.

Administrative and National Policy Requirements

Build America, Buy America

As expressed in Executive Order 14005, [Ensuring the Future is Made in All of America by All of America's Workers](#) (86 FR 7475), it is the policy of the executive branch to use terms and conditions of Federal financial assistance awards to maximize, consistent with law, the use of goods, products, and materials produced in, and services offered in, the United States.

Consistent with the requirements of the Build America, Buy America Act (Pub. L. 117-58, Division G, Title IX, Subtitle A, November 15, 2021), no funding made available through this funding opportunity may be obligated for an award unless all iron, steel, manufactured products, and construction materials used in the project are produced in the United States. For additional information, visit NSF's [Build America, Buy America](#) webpage.

Special Award Conditions:

Attribution of support in publications must acknowledge the joint program, as well as the funding organization and award number, by including the phrase, "as part of the Joint DMS/NIGMS Initiative to Support Research at the Interface of the Biological and Mathematical Sciences."

Grants made by NSF will be subject to NSF's award conditions. Grants made by NIH will be subject to NIH's award conditions (see <http://grants.nih.gov/grants/policy/awardconditions.htm>).

C. Reporting Requirements

For all multi-year grants (including both standard and continuing grants), the Principal Investigator must submit an annual project report to the cognizant Program Officer no later than 90 days prior to the end of the current budget period. (Some programs or awards require submission of more frequent project reports). No later than 120 days following expiration of a grant, the PI also is required to submit a final project report, and a project outcomes report for the general public.

Failure to provide the required annual or final project reports, or the project outcomes report, will delay NSF review and processing of any future funding increments as well as any pending proposals for all identified PIs and co-PIs on a given award. PIs should examine the formats of the required reports in advance to assure availability of required data.

PIs are required to use NSF's electronic project-reporting system, available through Research.gov, for preparation and submission of annual and final project reports. Such reports provide information on accomplishments, project participants (individual and organizational), publications, and other specific products and impacts of the project. Submission of the report via Research.gov constitutes certification by the PI that the contents of the report are accurate and complete. The project outcomes report also must be prepared and submitted using Research.gov. This report serves as a brief summary, prepared specifically for the public, of the nature and outcomes of the project. This report will be posted on the NSF website exactly as it is submitted by the PI.

More comprehensive information on NSF Reporting Requirements and other important information on the administration of NSF awards is contained in the *NSF Proposal & Award Policies & Procedures Guide* (PAPPG) Chapter VII, available electronically on the NSF Website at https://www.nsf.gov/publications/pub_summ.jsp?ods_key=pappg.

Grants made by NSF will be subject to NSF's reporting requirements. Grants made by NIH will be subject to NIH's reporting requirements.

VIII. Agency Contacts

Please note that the program contact information is current at the time of publishing. See program website for any updates to the points of contact.

General inquiries regarding this program should be made to:

- Zhilan J. Feng, Program Director, NSF/DMS, telephone: (703) 292-7523, email: dms-nigms@nsf.gov
- Vu Dinh, Program Director, NSF/DMS, telephone: (703) 292-4306, email: vdinh@nsf.gov
- John E. Kolassa, Program Director, telephone: (703) 292-4733, email: jkolassa@nsf.gov
- Tapabrata Maiti, Program Director, telephone: (703) 292-5307, email: tmaiti@nsf.gov
- Peggy Wang, Program Director, telephone: 301-594-0800, email: peggy.wang@nih.gov
- Han Nguyen, Program Director, NIH/NIGMS, telephone: (301) 496-7301, email: han.nguyen@nih.gov

For questions related to the use of NSF systems contact:

- NSF Help Desk: 1-800-381-1532
- Research.gov Help Desk e-mail: rgov@nsf.gov

For questions relating to Grants.gov contact:

- Grants.gov Contact Center: If the Authorized Organizational Representatives (AOR) has not received a confirmation message from Grants.gov within 48 hours of submission of application, please contact via telephone: 1-800-518-4726; e-mail: support@grants.gov.

IX. Other Information

The NSF website provides the most comprehensive source of information on NSF Directorates (including contact information), programs and funding opportunities. Use of this website by potential proposers is strongly encouraged. In addition, "NSF Update" is an information-delivery system designed to keep potential proposers and other interested parties apprised of new NSF funding opportunities and publications, important changes in proposal and award policies and procedures, and upcoming NSF [Grants Conferences](#). Subscribers are informed through e-mail or the user's Web browser each time new publications are issued that match their identified interests. "NSF Update" also is available on [NSF's website](#).

Grants.gov provides an additional electronic capability to search for Federal government-wide grant opportunities. NSF funding opportunities may be accessed via this mechanism. Further information on Grants.gov may be obtained at <https://www.grants.gov>.

About the National Institute of General Medical Sciences

The National Institute of General Medical Sciences (NIGMS) supports basic research that increases our understanding of biological processes and lays the foundation for advances in disease diagnosis, treatment, and prevention. NIGMS-funded scientists investigate how living systems work at a range of levels from molecules and cells to tissues and organs, in research organisms, humans, and populations. Additionally, to ensure the vitality and continued productivity of the research enterprise, NIGMS provides leadership in training future scientists, and in developing research capacity throughout the country.

For more information, go to the NIGMS website at <http://www.nigms.nih.gov/>.

The National Science Foundation

The National Science Foundation (NSF) is an independent Federal agency created by the National Science Foundation Act of 1950, as amended (42 USC 1861-75). The Act states the purpose of the NSF is "to promote the progress of science; [and] to advance the national health, prosperity, and welfare by supporting research and education in all fields of science and engineering."

NSF funds research and education in most fields of science and engineering. It does this through grants and cooperative agreements to more than 2,000 colleges, universities, K-12 school systems, businesses, informal science organizations and other research organizations throughout the US. The Foundation accounts for about one-fourth of Federal support to academic institutions for basic research.

NSF receives approximately 55,000 proposals each year for research, education and training projects, of which approximately 11,000 are funded. In addition, the Foundation receives several thousand applications for graduate and postdoctoral fellowships. The agency operates no laboratories itself but does support National Research Centers, user facilities, certain oceanographic vessels and Arctic and Antarctic research stations. The Foundation also supports cooperative research between universities and industry, US participation in international scientific and engineering efforts, and educational activities at every academic level.

Facilitation Awards for Scientists and Engineers with Disabilities (FASSED) provide funding for special assistance or equipment to enable persons with disabilities to work on NSF-supported projects. See the *NSF Proposal & Award Policies & Procedures Guide* Chapter II.E.6 for instructions regarding preparation of these types of proposals.

The National Science Foundation has Telephonic Device for the Deaf (TDD) and Federal Information Relay Service (FIRS) capabilities that enable individuals with hearing impairments to communicate with the Foundation about NSF programs, employment or general information. TDD may be accessed at (703) 292-5090 and (800) 281-8749, FIRS at (800) 877-8339.

The National Science Foundation Information Center may be reached at (703) 292-5111.

The National Science Foundation promotes and advances scientific progress in the United States by competitively awarding grants and cooperative agreements for research and education in the sciences, mathematics, and engineering.

To get the latest information about program deadlines, to download copies of NSF publications, and to access abstracts of awards, visit the NSF Website at <https://www.nsf.gov>

- **Location:** 2415 Eisenhower Avenue, Alexandria, VA 22314
- **For General Information** (NSF Information Center): (703) 292-5111
- **TDD (for the hearing-impaired):** (703) 292-5090
- **To Order Publications or Forms:**
 - Send an e-mail to: nsfpubs@nsf.gov
 - or telephone: (703) 292-8134
- **To Locate NSF Employees:** (703) 292-5111

Privacy Act And Public Burden Statements

The information requested on proposal forms and project reports is solicited under the authority of the National Science Foundation Act of 1950, as amended. The information on proposal forms will be used in connection with the selection of

qualified proposals; and project reports submitted by awardees will be used for program evaluation and reporting within the Executive Branch and to Congress. The information requested may be disclosed to qualified reviewers and staff assistants as part of the proposal review process; to proposer institutions/grantees to provide or obtain data regarding the proposal review process, award decisions, or the administration of awards; to government contractors, experts, volunteers and researchers and educators as necessary to complete assigned work; to other government agencies or other entities needing information regarding applicants or nominees as part of a joint application review process, or in order to coordinate programs or policy; and to another Federal agency, court, or party in a court or Federal administrative proceeding if the government is a party. Information about Principal Investigators may be added to the Reviewer file and used to select potential candidates to serve as peer reviewers or advisory committee members. See [System of Record Notices](#), NSF-50, "Principal Investigator/Proposal File and Associated Records," and NSF-51, "Reviewer/Proposal File and Associated Records." Submission of the information is voluntary. Failure to provide full and complete information, however, may reduce the possibility of receiving an award.

An agency may not conduct or sponsor, and a person is not required to respond to, an information collection unless it displays a valid Office of Management and Budget (OMB) control number. The OMB control number for this collection is 3145-0058. Public reporting burden for this collection of information is estimated to average 120 hours per response, including the time for reviewing instructions. Send comments regarding the burden estimate and any other aspect of this collection of information, including suggestions for reducing this burden, to:

Suzanne H. Plimpton
Reports Clearance Officer
Policy Office, Division of Institution and Award Support
Office of Budget, Finance, and Award Management
National Science Foundation
Alexandria, VA 22314

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